

Mocking Dependencies in Spring Tests using Mockito

Exercise 1: Mocking a Service Dependency in a Controller Test

```
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.test.web.servlet.MockMvc;

import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.*;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

@WebMvcTest(UserController.class)
public class UserControllerTest {

    @Autowired
    private MockMvc mockMvc;

    @MockBean
    private UserService userService;

    @Test
    public void testGetUser() throws Exception {
        User user = new User();
        User user = new User();
        user.setId(1L);
        user.setName("Vikram Nair");

        when(userService.getUserById(1L)).thenReturn(user);

        mockMvc.perform(get("/users/1"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.name").value("Vikram Nair"));
    }
}
```

Output:

```
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 6.619 s -- in
com.example.demo.UserControllerTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.037 s -- in
com.example.demo.UserIntegrationTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.285 s -- in
com.example.demo.UserServiceTest

Total: 5 tests run, 0 failures, 0 errors, 0 skipped
```

Exercise 2: Mocking a Repository in a Service Test

```
import org.junit.jupiter.api.Test;
import org.junit.jupiter.api.extension.ExtendWith;
import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.junit.jupiter.MockitoExtension;

import java.util.Optional;
import static org.junit.jupiter.api.Assertions.*;
import static org.mockito.Mockito.*;

@ExtendWith(MockitoExtension.class)
public class UserServiceTest {

    @Mock
    private UserRepository userRepository;

    @InjectMocks
    private UserService userService;

    @Test
    public void testGetUserById() {
        User mockUser = new User();
        mockUser.setId(2L);
        mockUser.setName("Anjali Gupta");

        when(userRepository.findById(2L)).thenReturn(Optional.of(mockUser));

        User result = userService.getUserById(2L);

        assertNotNull(result);
        assertEquals("Anjali Gupta", result.getName());
    }
}
```

Output

```
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 6.544 s -- in
com.example.demo.UserControllerTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.668 s -- in
com.example.demo.UserIntegrationTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.260 s -- in
com.example.demo.UserServiceTest

Total: 4 tests run, 0 failures, 0 errors, 0 skipped
```

Exercise 3: Mocking a Service Dependency in an Integration Test

```
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.test.web.servlet.MockMvc;

import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.*;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

@SpringBootTest
@AutoConfigureMockMvc
public class UserIntegrationTest {

    @Autowired
    private MockMvc mockMvc;

    @MockBean
    private UserService userService;

    @Test
    @Test
    public void testGetUserEndpoint() throws Exception {
        User user = new User();
        user.setId(3L);
        user.setName("Rohan Desai");

        when(userService.getUserById(3L)).thenReturn(user);

        mockMvc.perform(get("/users/3"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.name").value("Rohan Desai"));
    }
}
```

Output:

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 6.091 s -- in
com.example.demo.UserControllerTest

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.734 s -- in
com.example.demo.UserIntegrationTest

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.187 s -- in
com.example.demo.UserServiceTest

Total: 3 tests run, 0 failures, 0 errors, 0 skipped